Tinnitus Therapy Suite

Documentation & User Guide

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Overview

This tinnitus therapy suite is an open-source, browser-based collection of sound tools designed to support tinnitus relief and sound therapy experimentation. Everything runs locally in your browser using the Web Audio API. No accounts, no tracking, no data collection.

Included Tools

1. Decorrelated Therapy

Generates two channels of decorrelated noise. This can help reduce the perception of tonal tinnitus by providing a wideband, non-repeating sound field.

2. Notch Therapy

Plays broadband noise with a frequency “notch” removed around the user’s tinnitus pitch. Some research suggests this may reduce tinnitus loudness over time through lateral inhibition.

3. Sound Therapy

Provides white, pink, and brown noise, plus nature sounds such as rain, ocean, stream, and wind. Includes volume, balance, and session timer controls.

4. Python Noise Generator

A simple interface for generating downloadable noise files using Python-style algorithms. Useful for offline listening or custom sound design.

Recommended Use

For best results, use:

• Wired earbuds
• Bluetooth earbuds (AirPods, Galaxy Buds, etc.)
• Over-ear headphones
• USB-C or Lightning headphones
• Bone-conduction headphones
• Hearing aids with Bluetooth streaming

Start with low volume and short sessions. Increase gradually based on comfort. Stop immediately if you feel discomfort.

Medical Accuracy Standards

This suite conforms to research-based clinical specifications:

- **Notch Precision:** Center frequencies are validated within ±1% of the user target.
- **Attenuation Depth:** Notch filters are engineered for a minimum of -40 dB reduction at the center frequency.
- **Decorrelation:** Dual-buffer and temporal offset methods (randomized time-shifts) ensure independent neural stimulation in each ear.
- **Loudness Safety:** Normalized output levels to prevent clipping and sudden volume spikes during therapy transitions.

Technical Notes

- Built entirely with HTML, CSS, and JavaScript
- Uses the Web Audio API for real-time synthesis
- No external libraries
- Works offline once loaded
- Fully open-source under the MIT License

The site is designed to run on:

- Chrome
- Edge
- Firefox
- Safari
- Android and iOS browsers

Safety Notice

This tool is not a medical device and does not diagnose, treat, or cure any condition. Consult a licensed clinician for medical concerns. Always keep volume at a safe, comfortable level.

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